Leader: Hồ Nguyễn Phú
- Cao Tuệ Anh
- Phạm Trọng Quý
- Châu Minh Quân
- Hồng Nguyễn Phúc

Phu
Hồ Nguyễn Phú

Tutorial 7

1)

a. An office has 4 copying machines and the random variable X measures how many of them are in use at a particular time.

So these probability values must satisfy:

$$\sum_{i=0}^{4} p_i = 1$$

$$P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) = 1$$

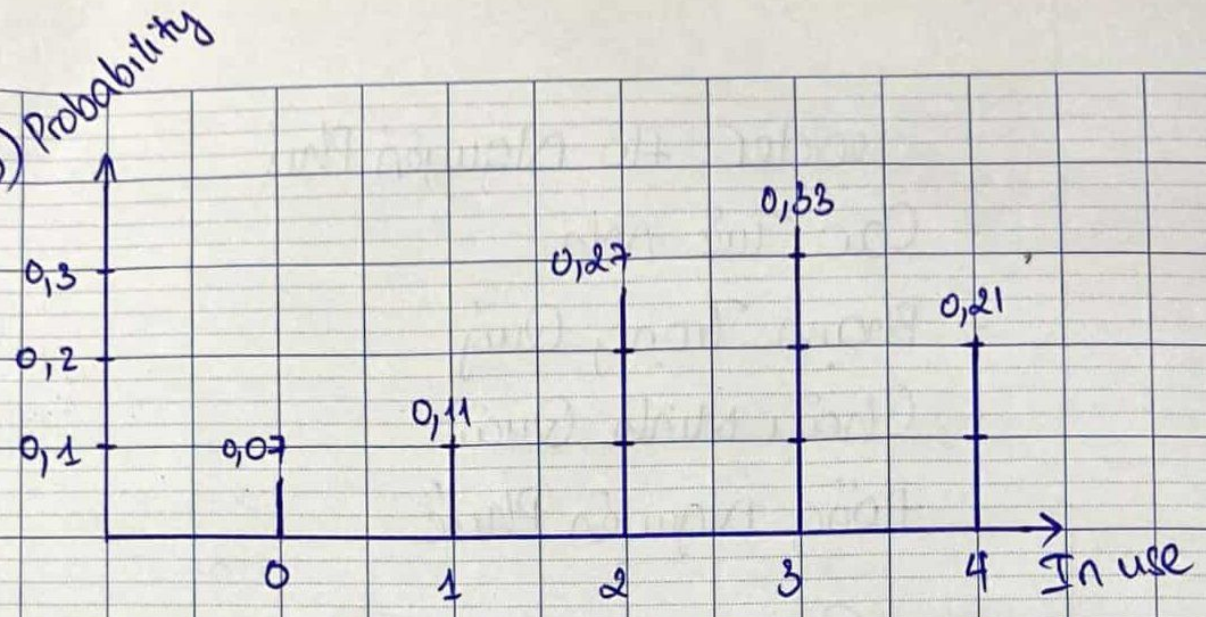
$$\Leftrightarrow 0{,}08 + 0{,}11 + 0{,}27 + 0{,}33 + P(X=4) = 1$$

$$\Leftrightarrow 0{,}79 + P(X=4) = 1$$

$$\Leftrightarrow P(X=4) = 1 - 0{,}79 = 0{,}21$$

Phu
Hồ Nguyễn
Phú

b)



The line graph of the probability mass function

c)

x_i	0	1	2	3	4
p_i	0,08	0,11	0,27	0,33	0,21

$-\infty < x < 0 \Rightarrow F(x) = P(\text{machine in use} \leq x) = 0$

$0 \leq x < 1 \Rightarrow F(x) = P(\text{machine in use} \leq x) = 0{,}08$

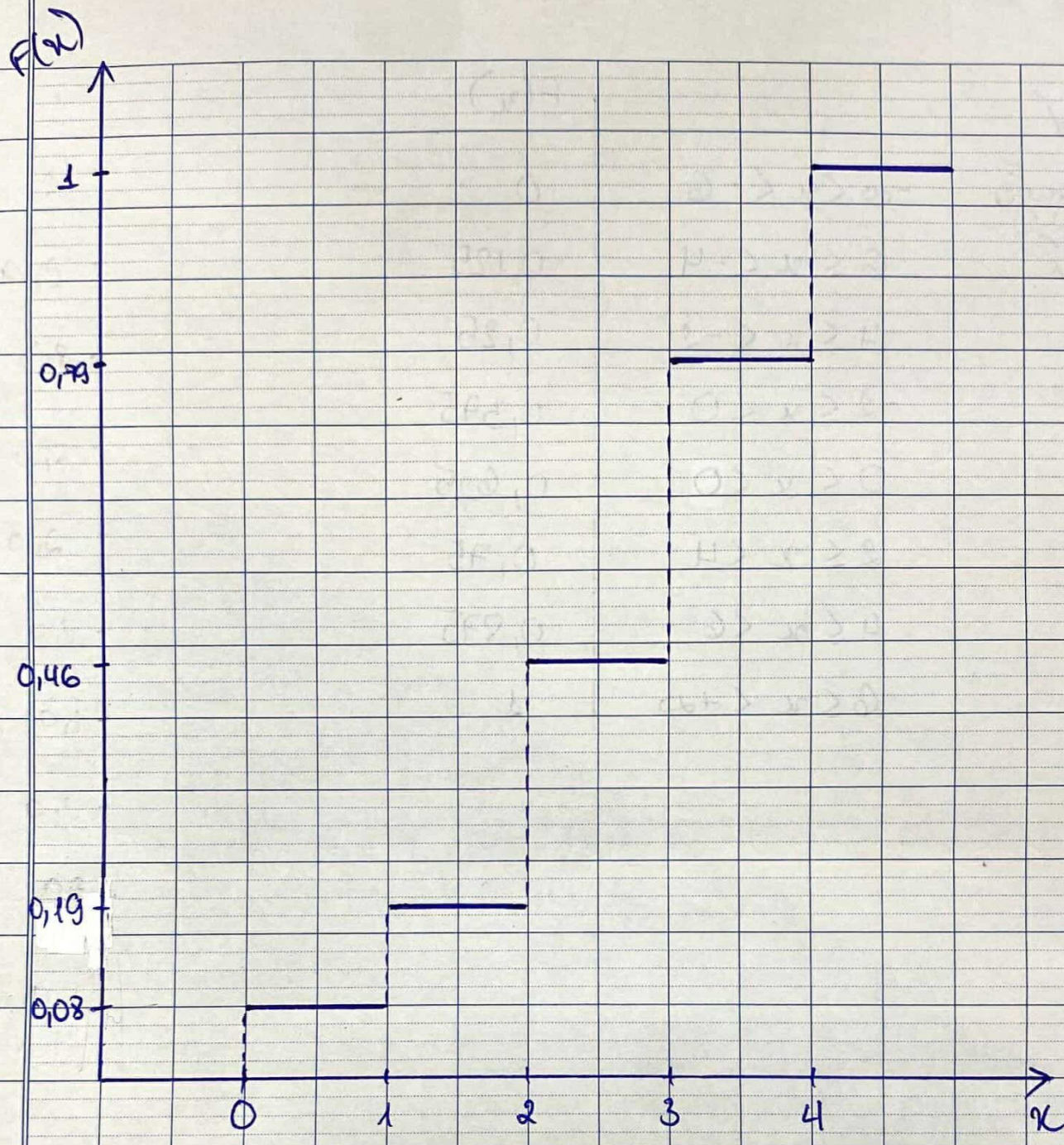
$1 \leq x < 2 \Rightarrow F(x) = P(\text{machine in use} \leq x) = 0{,}08 + 0{,}11$
$= 0{,}19$

$2 \leq x < 3 \Rightarrow F(x) = P(\text{machine in use} \leq x) = 0{,}08 + 0{,}11 + 0{,}27$
$= 0{,}46$

$3 \leq x < 4 \Rightarrow F(x) = P(\text{machine in use} \leq x) = 0{,}08 + 0{,}11 + 0{,}27 + 0{,}33$
$= 0{,}79$

$4 \leq x < +\infty \Rightarrow F(x) = P(\text{machine in use} \leq x) = 0{,}08 + 0{,}11 + 0{,}27 + 0{,}33 + 0{,}2$
$= 1$

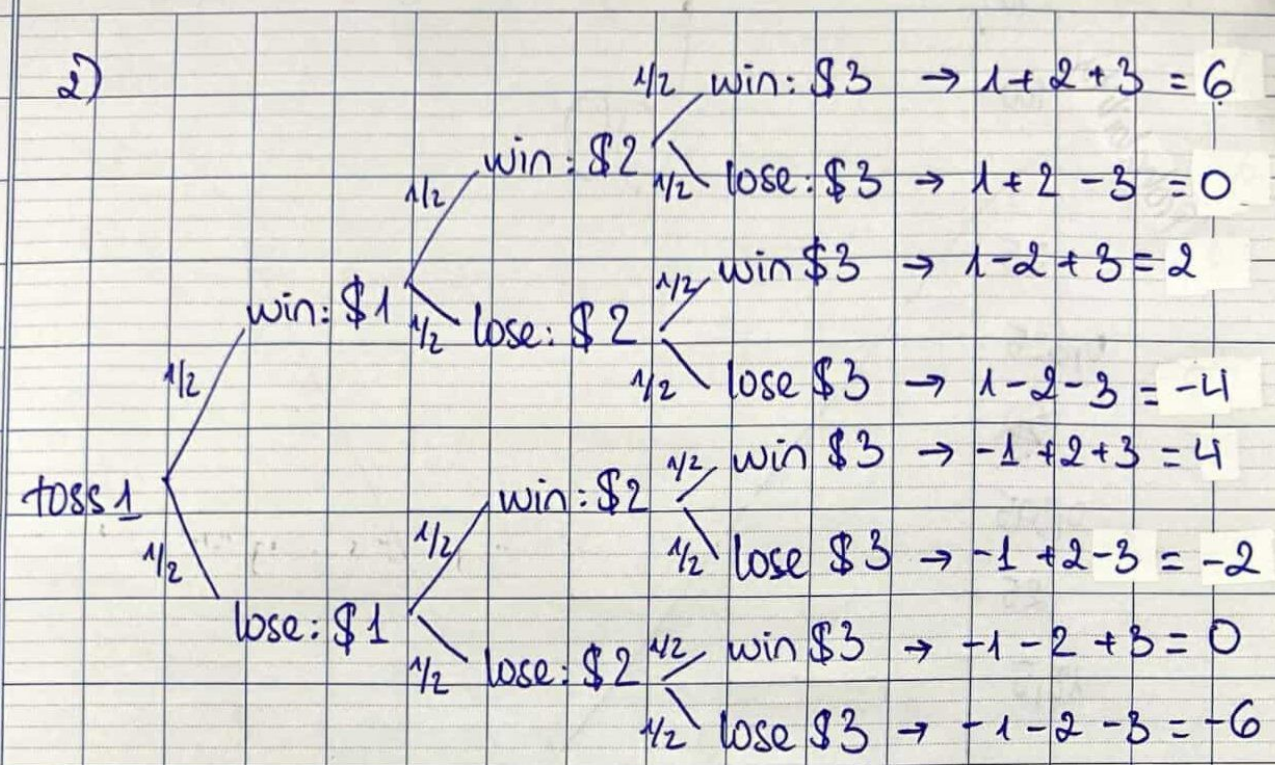
Phu
Hồ Nguyễn
Phú



The graph of the cumulative distribution function

Phu
Hồ Nguyễn
Phú

2)



a)

x_i	−6	−4	−2	0	2	4	6
p_i	1/8	1/8	1/8	2/8	1/8	1/8	1/8

b) $-\infty < x < -6 \Rightarrow F(x) = P(\text{total winnings} \le x) = 0$

$-6 \le x < -4 \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 = 0{,}125$

$-4 \le x < -2 \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 + 1/8 = 0{,}25$

$-2 \le x < 0 \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 + 1/8 + 1/8 = 0{,}375$

$0 \le x < 2 \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 + 1/8 + 1/8 + 2/8 = 0{,}625$

$2 \le x < 4 \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 + 1/8 + 1/8 + 2/8 + 1/8$
$= 0{,}75$

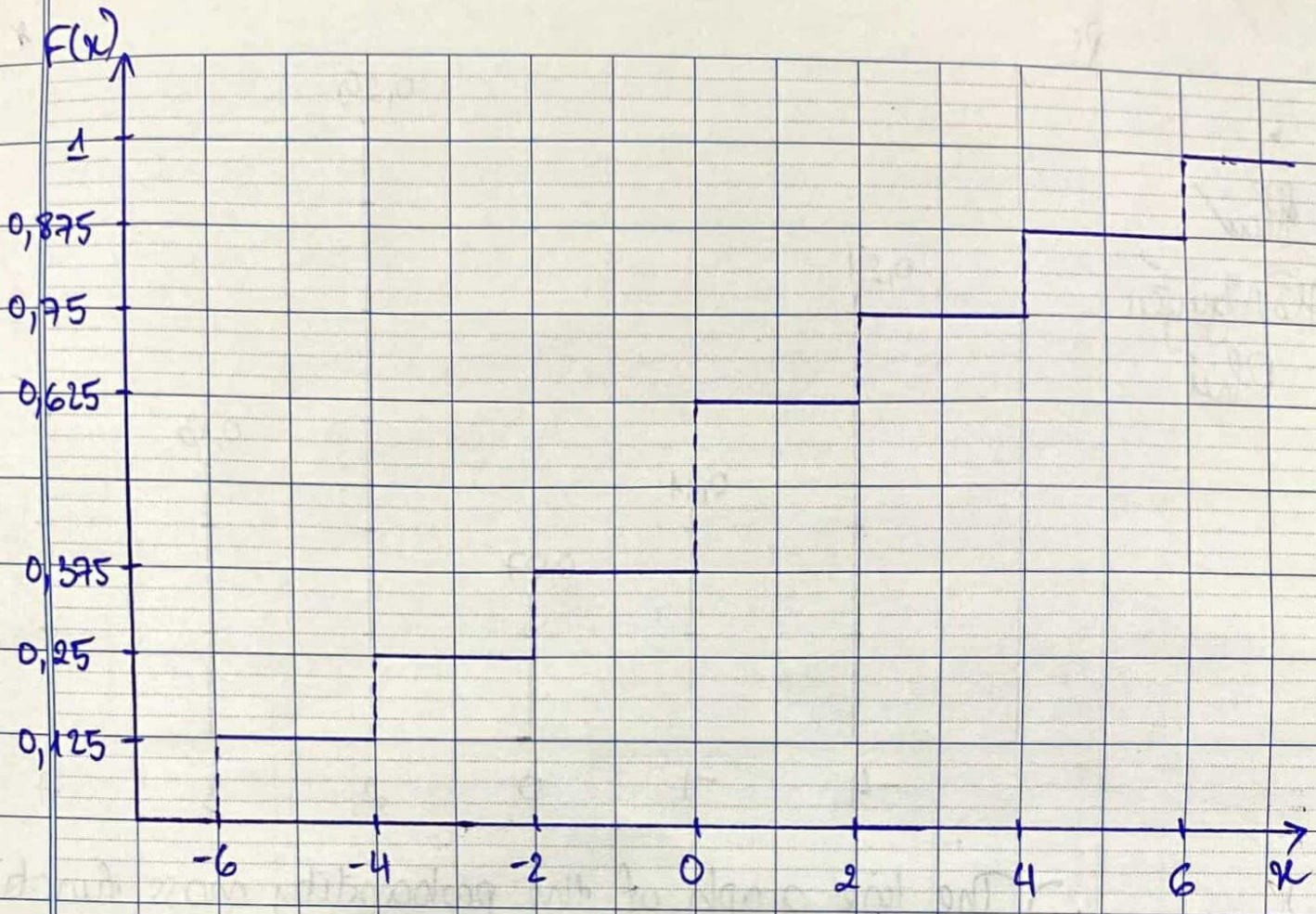
$4 \le x < 6 \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 + 1/8 + 1/8 + 2/8 + 1/8 + 1/8$
$= 0{,}875$

$6 \le x < +\infty \Rightarrow F(x) = P(\text{total winnings} \le x) = 1/8 + 1/8 + 1/8 + 2/8 + 1/8 + 1/8 + 1/8$
$= 1$

Phu
Hồ Nguyễn
Phú

	$F(x)$
$-\infty < x < -6$	0
$-6 \le x < -4$	0,125
$-4 \le x < -2$	0,25
$-2 \le x < 0$	0,375
$0 \le x < 2$	0,625
$2 \le x < 4$	0,75
$4 \le x < 6$	0,875
$6 \le x < +\infty$	1

Phu
Hồ
Nguyên
Phú



The graph of the cumulative distribution function

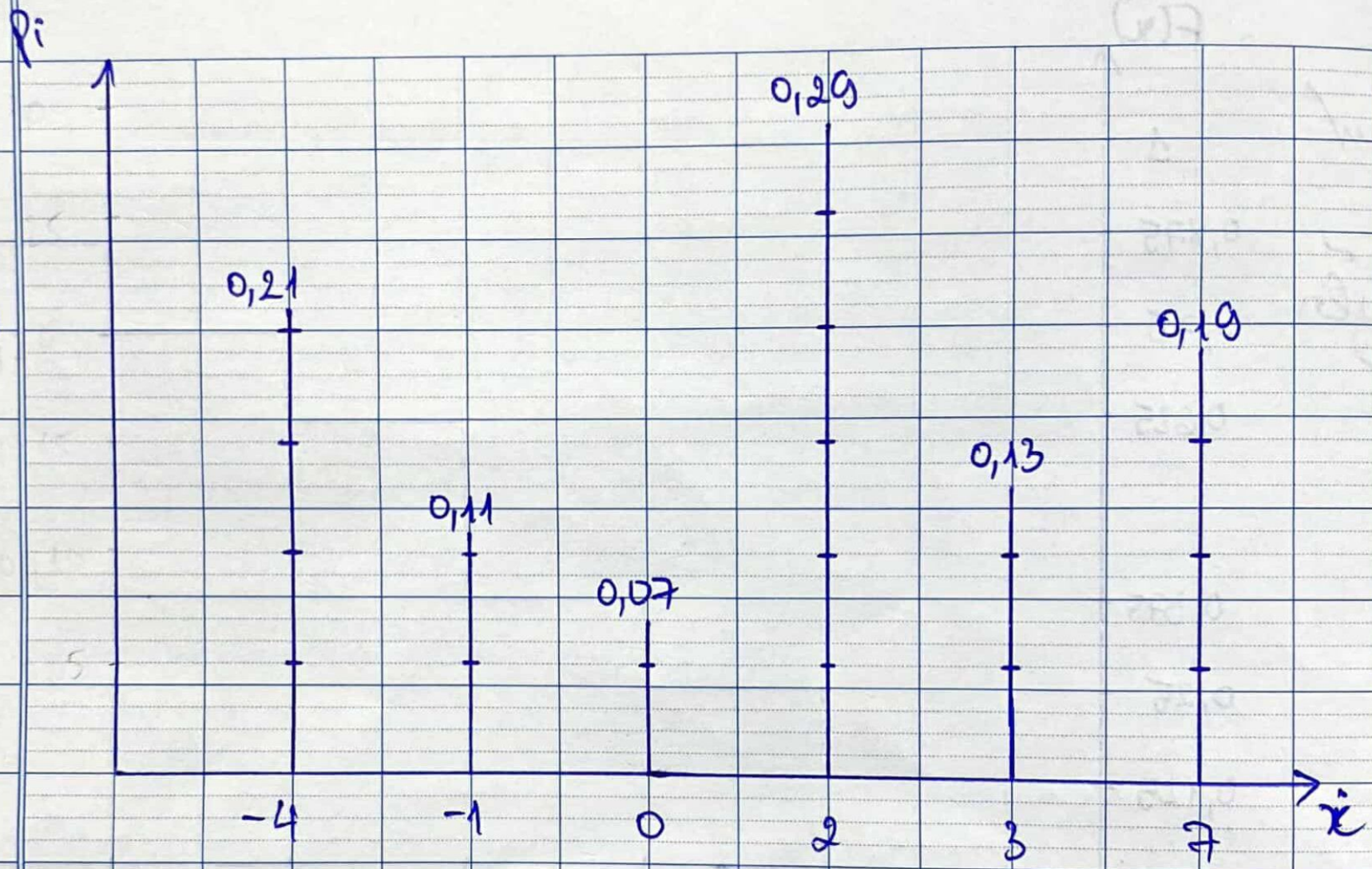
c)

The most likely value of random variable X is $X=0$ with $P(X=0)=0,25$ which is larger than the rest (only 0,125)

3)

x_i	-4	-1	0	2	3	7
p_i	0,21	0,11	0,07	0,29	0,13	0,19

Phu
HỒ Nguyễn
Phú



The line graph of the probability mass function